

AI Room Configuration — Project Overview

This document describes repository structure, AI models, external services, API routes, and how to think about cost. For environment variables, see `.env.example` in the project root.

1. Repository layout

Path	Role
Root (<code>package.json</code>)	Next.js 14 application. Server-side AI runs in <code>app/api/*</code> .
<code>Internalconfigf/</code>	Vite + React UI (e.g. port 5173). Calls the Next app (typically port 3000) for <code>/api/*</code> .
<code>app/</code>	Pages (<code>app/page.tsx</code>), components, Zustand store, room editor libraries, server utilities.
<code>supabase/migrations/</code>	SQL migrations for catalog data (tiles, sofas, lighting, chairs, tables, decor, etc.).
<code>lift subject/</code>	Separate Vite-based experiment; uses its own Gemini env vars (<code>VITE_*</code>). Not the main Next deployment path.
<code>scripts/</code>	Helper scripts (e.g. seed generation).
<code>vercel.json</code>	API route maxDuration 300s for <code>app/api/**/*.*ts</code> .

Primary stack (production path): Next.js API + React UI (main app and/or Internalconfigf as client).

2. Dependencies (root `package.json`)

Package	Use
<code>next</code> , <code>react</code> , <code>react-dom</code>	App framework and UI.
<code>@runwayml/sdk</code>	Room tour video (<code>/api/room-tour-video</code>).
<code>@supabase/supabase-js</code>	Database and storage (catalogs, tour temp uploads).
<code>sharp</code>	Image resize, compositing, JPEG/PNG processing server-side.
<code>undici</code>	HTTP client usage where applicable.
<code>zustand</code>	Client state.

Internalconfigf has its own `package.json` (Vite, MUI, Radix, etc.) — UI only; AI is still on Next APIs when wired that way.

3. Environment variables (summary)

Variable	Required for	Notes
IMAGE_GENERATION_API_KEY	Gemini on Next server	Without it, /api/generate can return placeholder behavior (see route implementation).
GEMINI_TEXT_MODEL	Text / vision-text calls	Default in code: gemini-2.5-flash .
GEMINI_IMAGE_MODEL	Image generation & inpaint	Default: gemini-3-pro-image-preview .
GEMINI_IMAGE_TIMEOUT_MS	Inpaint / heavy image calls	Optional; max wait for Gemini image (e.g. 60s–600s).
RUNWAYML_API_SECRET	/api/room-tour-video	Runway Gen-4 Turbo image-to-video.
ROOM_TOUR_STORAGE_BUCKET	Tour video flow	Supabase bucket for temporary first-frame JPEG.
SUPABASE_SERVICE_ROLE_KEY	Server uploads / privileged ops	Keep server-only.
NEXT_PUBLIC_SUPABASE_URL , NEXT_PUBLIC_SUPABASE_ANON_KEY	Client + server Supabase access	Public anon key is exposed to browser.
SEGMENTATION_API_KEY , SEGMENTATION_API_URL	/api/segment	Your SAM-compatible endpoint.
GROUNDING_DINO_API_KEY , GROUNDING_DINO_API_URL	/api/detect-objects	Your detection endpoint.
RUNWAY_VIDEO_DURATION , RUNWAY_VIDEO_RATIO , RUNWAY_VIDEO_SEED	Optional Runway tunables	Defaults: 10s, 1280:720 .

Optional / legacy mentions in .env.example : STABILITY_AI_API_KEY , REPLICATE_API_TOKEN — not the primary path for the main generate flow (Gemini + Imagen fallback in code).

4. AI models and where they are used

4.1 Google Gemini (primary)

Model ID (typical)	Env override	Used for
gemini-2.5-flash	GEMINI_TEXT_MODEL	Prompt structuring, validation, component detection text, analysis routes.

Model ID (typical)	Env override	Used for
gemini-3-pro-image-preview	GEMINI_IMAGE_MODEL	Full room image generation; masked inpaint (add / edit / replace / erase) via <code>app/lib/server/geminiInpaintMask.ts</code> .
imagen-4.0-generate-001	(hardcoded fallback in <code>app/api/generate/route.ts</code>)	Fallback if primary image model fails.

API base: `generativelanguage.googleapis.com` (`/v1` or `/v1beta` depending on call).

4.2 Runway

Field	Value
SDK	<code>@runwayml/sdk</code>
Model	<code>gen4_turbo</code> (image-to-video)
Route	<code>POST/GET /api/room-tour-video</code>

4.3 Optional external HTTP APIs (you host or subscribe)

- **Segmentation:** `/api/segment` → `SEGMENTATION_API_URL` (SAM-style masks).
- **Object detection:** `/api/detect-objects` → `GROUNDING_DINO_API_URL` .

5. Next.js API routes (quick map)

Route	Primary backend
<code>/api/generate</code>	Gemini text + Gemini image; Imagen fallback; highest volume for “full room” generations.
<code>/api/add</code> , <code>/api/edit</code> , <code>/api/replace</code> , <code>/api/erase</code>	Gemini image (masked inpaint) + <code>sharp</code> compositing.
<code>/api/edit-cutout</code>	Gemini + image processing (see file).
<code>/api/validate-image-type</code>	Gemini text.
<code>/api/detect-components</code>	Gemini text.
<code>/api/detect-objects</code>	External detection API.
<code>/api/segment</code>	External segmentation API.
<code>/api/analyze-room</code> , <code>/api/analyze-external</code>	Room/property analysis (Gemini in route).
<code>/api/room-tour-video</code>	Runway + Supabase signed URL for first frame.

Route	Primary backend
<code>/api/product-variations</code> , <code>/api/mytypes-vitrified-tiles</code>	Supabase / catalog reads.

6. Cost guidance (usage-based; not a fixed project fee)

Exact spend depends on:

- Google billing tier, free credits, and **per-model** pricing.
- Runway plan and per-generation charges.
- Supabase plan and storage/egress.
- Your third-party segmentation/detection provider.

6.1 How to estimate Gemini

- Each full `/api/generate` **success path** often involves **at least one text generation** and **one image generation** (plus possible retries or Imagen fallback).
- Each **add/replace/erase/edit inpaint** is typically **one or more image-model calls** per user action (retries may add more).

Official pricing: [Gemini API pricing](#) — match the **exact model strings** you set in `GEMINI_TEXT_MODEL` and `GEMINI_IMAGE_MODEL` .

6.2 Runway

- Charged **per video job** (model, duration, resolution).
- **Reference:** [Runway pricing](#) for Gen-4 / Turbo.

6.3 Supabase

- Generally smaller than AI at moderate scale unless you store large media or have high egress.

6.4 Simple monthly formula (fill in your averages)

```
Monthly estimate ≈
(room_generations × cost_per_generate)
+ (inpaint_operations × cost_per_inpaint)
+ (tour_videos × cost_per_video)
+ Supabase_subscription
+ segmentation_detection_calls × provider_unit_price
```

Export usage from **Google Cloud / AI Studio billing**, **Runway dashboard**, and **Supabase** usage reports to calibrate `cost_per_*` .

7. Security

- Never commit `.env.local` or real API keys.
- Rotate any key that was ever pasted into chat, docs, or screenshots.
- Restrict `SUPABASE_SERVICE_ROLE_KEY` to server-only code.

8. Related files

File	Content
<code>.env.example</code>	Variable names and comments.
<code>app/api/generate/route.ts</code>	Main generation pipeline, model names, timeouts, fallbacks.
<code>app/lib/server/geminiInpaintMask.ts</code>	Inpaint model ID and timeout helper.
<code>app/api/room-tour-video/route.ts</code>	Runway <code>gen4_turbo</code> , duration/ratio parsing.

Last updated from codebase scan (structure and env names may drift; verify against source if something fails after an upgrade.)